



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to JP AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

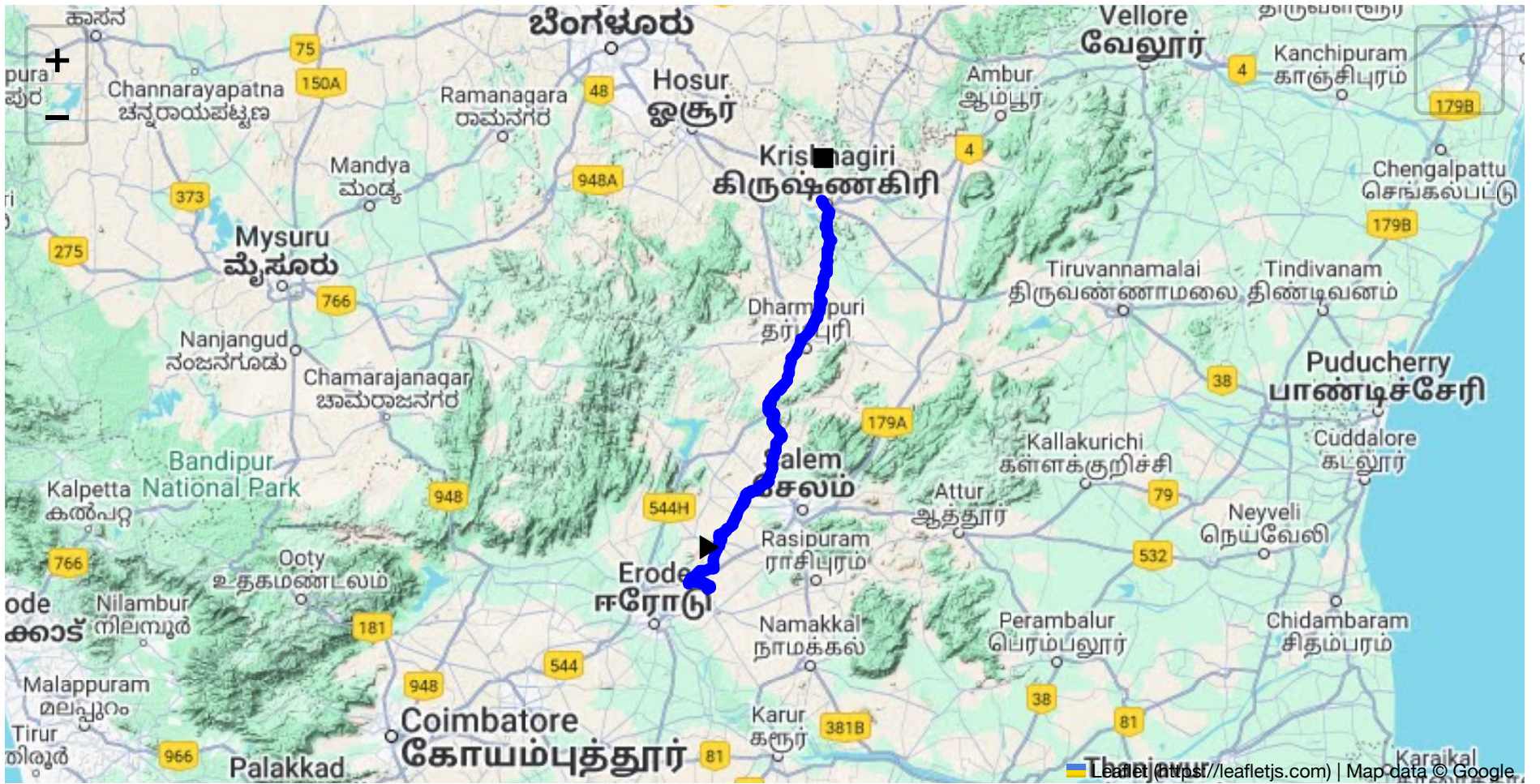
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 154.00 km
Estimated Duration: 2.9 hours
Adjusted Duration (Heavy Vehicle): 3.7 hours

Start: (11.4381, 77.8734)
End: (12.53901, 78.202517)



Welcome to the Journey Risk Management Study

Route Overview

The route begins in Sangagiri, Tamil Nadu, and ends in Krishnagiri, Tamil Nadu, covering approximately 154 kilometers. It primarily follows state highways and passes through key localities such as Bhavani Main Road and SH 86. The journey typically takes about 2.93 hours for heavy vehicles, including those carrying hazardous materials.

Weather Conditions

- **Typical Weather Patterns:** Tamil Nadu generally experiences hot and humid weather with monsoon rains from October to December.
- **Potential Hazards:** Heavy rains during the northeast monsoon can lead to waterlogging and decreased road visibility. The summer months might pose risks of overheating for vehicles.

Traffic Patterns

- **Peak Hours:** Expect increased traffic during morning (7:00 - 9:30 AM) and evening hours (5:00 - 8:00 PM), especially near towns.
- **Congestion-Prone Areas:** Bhavani and Krishnagiri often face congestion due to local traffic and commercial activities.

Road Quality and Infrastructure

- **General Assessment:** The route's state highways are moderately maintained. Potholes and uneven surfaces may be encountered, particularly post-monsoon.
- **Bridge and Flyover Conditions:** Regular checks are necessary for structural integrity due to wear and tear from heavy vehicles.

Alternative Routes

- In Case of Emergencies:** An alternative route can be through NH44 if available, although detours might increase travel time due to longer distances.

Local Regulations

- Hazardous Material Transport:** Mandatory permits and restricted movement during certain hours in populated areas. Ensure compliance with the Tamil Nadu Pollution Control Board guidelines.

Historical Incidents

- Notable Incidents:** Incidents involving heavy vehicles have been reported in the past, often due to mechanical failures or lack of adherence to safety standards. Always consider past accident zones as high-risk areas.

Environmental Considerations

- Sensitive Areas:** Some parts of the route may pass near agricultural lands where chemical spills could affect the environment.
- Waste Management:** Vehicles must ensure no leakage of hazardous materials to prevent soil and water contamination.

Communication Coverage

- Potential Dead Zones:** Rural stretches might have poor network coverage. Pre-download maps and important contacts, and consider satellite communication equipment for emergencies.

Emergency Response Times

- Estimation:** Response times can vary significantly. In urban areas or close to towns, 30-45 minutes. In rural stretches, expect up to an hour or more.

Overall Summary of Risk Assessment

- Summary:** The route poses moderate risks, mainly from weather conditions, traffic congestion, and the state of highways. Hazardous material transportation adds a layer of complexity, requiring strict adherence to regulations and preparedness for emergencies. Improved communication tools and contingency plans for detours can mitigate some risks. It is crucial for drivers to be trained in emergency response and familiar with the route dynamics, ensuring a safer journey for both cargo and personnel.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44900, 77.87406	15 KM/Hr
3	Turn	High	11.45353, 77.85697	15 KM/Hr
4	Turn	Medium	11.45781, 77.85203	30 KM/Hr
5	Turn	High	11.46307, 77.84941	15 KM/Hr
6	Turn	Medium	11.45546, 77.81800	30 KM/Hr
7	Blind Spot	Blind Spot	11.45605, 77.81683	10 KM/Hr
8	Blind Spot	Blind Spot	11.49400, 77.88538	10 KM/Hr
9	Turn	High	11.49469, 77.88488	15 KM/Hr
10	Turn	High	11.55614, 77.89764	15 KM/Hr
11	Turn	Medium	11.55680, 77.89969	30 KM/Hr
12	Turn	High	11.58218, 77.90845	15 KM/Hr
13	Turn	High	11.58330, 77.90642	15 KM/Hr
14	Turn	Medium	11.59189, 77.90645	30 KM/Hr
15	Turn	Medium	11.59328, 77.90809	30 KM/Hr
16	Turn	Medium	11.63867, 77.94841	30 KM/Hr
17	Turn	High	11.64217, 77.95200	15 KM/Hr
18	Turn	Medium	11.64712, 77.95118	30 KM/Hr
19	Turn	Medium	11.67858, 77.96834	30 KM/Hr
20	Turn	Medium	11.70062, 77.97961	30 KM/Hr
21	Turn	High	11.70218, 77.97866	15 KM/Hr
22	Turn	Medium	11.71399, 77.99580	30 KM/Hr
23	Turn	Medium	11.71917, 78.01123	30 KM/Hr
24	Turn	Medium	11.73769, 78.03451	30 KM/Hr
25	Turn	High	11.73651, 78.03871	15 KM/Hr
26	Turn	Medium	11.74271, 78.04717	30 KM/Hr
27	Turn	Medium	11.86847, 78.08367	30 KM/Hr
28	Turn	Medium	11.87942, 78.08197	30 KM/Hr
29	Turn	Medium	11.92585, 78.06507	30 KM/Hr
30	Turn	Medium	11.93704, 78.05402	30 KM/Hr
31	Turn	High	11.95818, 78.04978	15 KM/Hr
32	Turn	Medium	11.95988, 78.04948	30 KM/Hr
33	Turn	Medium	11.96304, 78.05136	30 KM/Hr
34	Turn	High	11.96399, 78.05135	15 KM/Hr
35	Turn	High	11.96580, 78.04923	15 KM/Hr
36	Turn	Medium	11.96675, 78.04939	30 KM/Hr
37	Turn	Medium	11.96791, 78.05070	30 KM/Hr
38	Turn	Medium	12.19881, 78.18157	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
39	Turn	Medium	12.22459, 78.19136	30 KM/Hr
40	Turn	Medium	12.25060, 78.19977	30 KM/Hr
41	Turn	Medium	12.25329, 78.20039	30 KM/Hr
42	Turn	Medium	12.29110, 78.19909	30 KM/Hr
43	Turn	Medium	12.39904, 78.21764	30 KM/Hr
44	Turn	Medium	12.40156, 78.21845	30 KM/Hr
45	Turn	Medium	12.40589, 78.22275	30 KM/Hr
46	Turn	Medium	12.45018, 78.21828	30 KM/Hr
47	Turn	Medium	12.50553, 78.22318	30 KM/Hr
48	Turn	Medium	12.53868, 78.20296	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
2	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
3	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
4	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
5	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
6	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
7	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
8	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
10	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
11	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
12	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
13	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
14	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
15	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
16	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
17	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
18	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
19	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
20	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
21	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
23	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
24	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
25	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
26	hospital	Government Hospital	12.3048002, 78.2060881	30 km/h	Medium
31	hospital	TCR Multi Speciality Hospital	12.5088183, 78.2237657	30 km/h	Medium
32	clinic	Shri Narayana Bone & Joint Clinic	12.523617, 78.210613	30 km/h	Medium
33	clinic	AZ Family Clinic	12.5234942, 78.2127422	30 km/h	Medium
35	hospital	KVS Hospital	12.5277857, 78.2055049	30 km/h	Medium
36	clinic	Shree Ramachandra ENT Centre	12.5290096, 78.2037359	30 km/h	Medium
37	hospital	Ga Heart Hospital	12.5287708, 78.2055336	30 km/h	Medium
38	clinic	Psychiatric Centre	12.529476, 78.205064	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
9	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
22	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
27	school	Government Boys Higher secondary school	12.3047059, 78.2065495	30 km/h	Medium
28	school	Nalanda Metric Higher Secondary School	12.488407, 78.21856	30 km/h	Medium
29	school	Nalanda International Public School	12.4994979, 78.2235094	30 km/h	Medium
30	school	Don Bosco School Ground	12.505954, 78.2222003	30 km/h	Medium
34	school	Vivekananda School	12.5231937, 78.2121429	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45781, 77.85203



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45546, 77.81800



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.45605, 77.81683



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49400, 77.88538



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49469, 77.88488



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.55614, 77.89764



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.58330, 77.90642



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59189, 77.90645



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59328, 77.90809



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63867, 77.94841



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64217, 77.95200



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64712, 77.95118



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67858, 77.96834



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.70062, 77.97961



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.70218, 77.97866



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71399, 77.99580



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Speed Limit: 30 KM/Hr
Coordinates: 11.71917, 78.01123



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Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73769, 78.03451



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Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73651, 78.03871



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74271, 78.04717



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Speed Limit: 30 KM/Hr
Coordinates: 11.86847, 78.08367



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87942, 78.08197



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.92585, 78.06507



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93704, 78.05402



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95818, 78.04978



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95988, 78.04948



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96304, 78.05136



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96399, 78.05135



Risk Type: Turn
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Speed Limit: 15 KM/Hr
Coordinates: 11.96580, 78.04923



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96675, 78.04939



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96791, 78.05070



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Coordinates: 12.25329, 78.20039



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.29110, 78.19909



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.39904, 78.21764



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.40156, 78.21845



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.40589, 78.22275



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.45018, 78.21828



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.50553, 78.22318



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.53868, 78.20296
